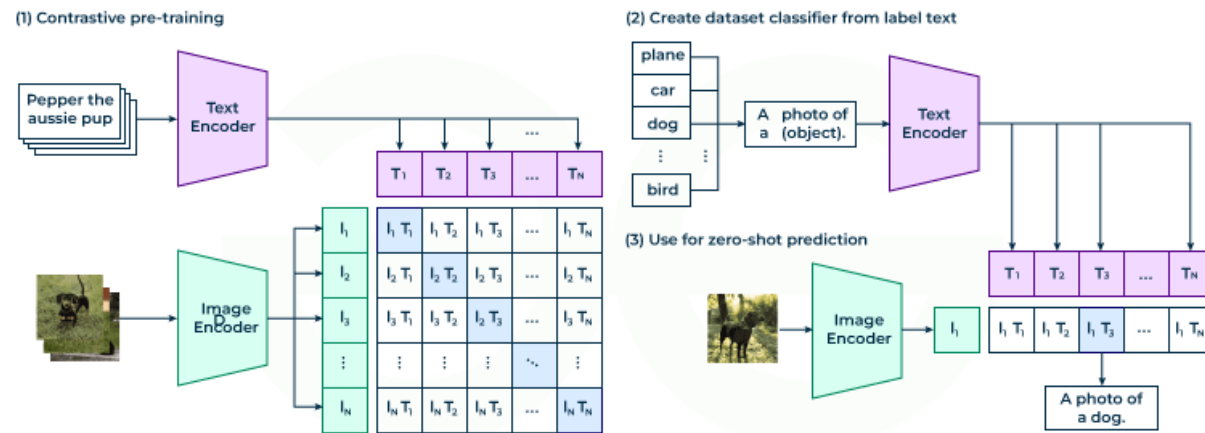


CLIP: The Bridge Between Pixels and Words



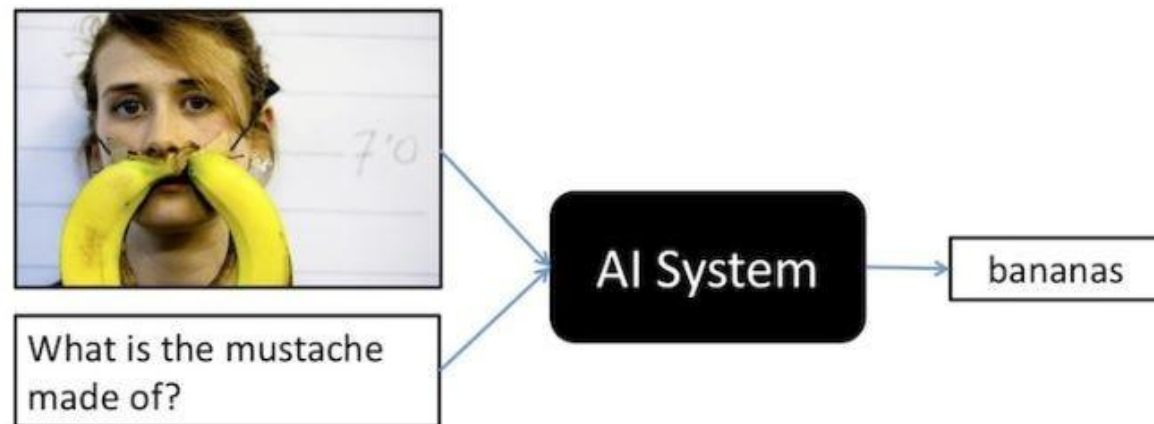
CLIP (Contrastive Language-Image Pretraining)



- Developed by OpenAI, **CLIP** teaches AI to understand images by reading their captions.
- It uses two encoders: one for images and one for text.
- **Contrastive Learning:** The model is trained to "pull" matching image-text pairs close together in mathematical space while "pushing" non-matching pairs apart.
- This allows for **Zero-Shot Classification**, where a model can identify an object it was never explicitly trained on simply by understanding the description.

VQA: Conversing with Images

- **VQA** allows models like GPT-4o or Llama-Vision to "see" an image and answer complex questions about it.
- It's not just tagging (e.g., "Dog"); it's reasoning (e.g., "Is the dog's leash tied to the bench?").
- Applications range from medical diagnostics (analyzing X-rays) to accessibility tools for the visually impaired.



Multimodal Transformers: Shared Intelligence

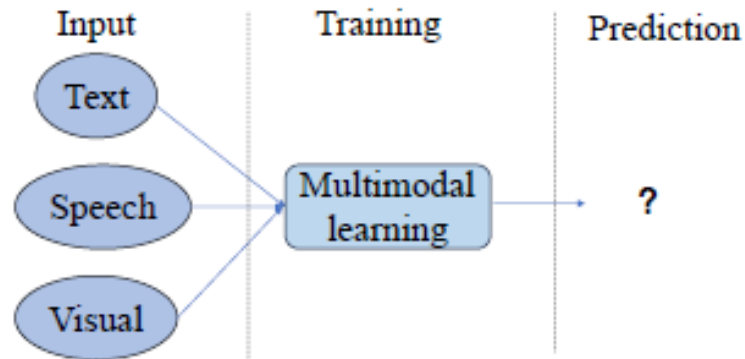


Fig. 1: The definition of multimodal.

- A **Multimodal Transformer** processes both pixels and text tokens in the same mathematical "engine."
- Images are broken into small "patches" (Visual Tokens), which the Transformer treats just like words in a sentence.
- This unified architecture allows the model to find deep cross-modal relationships that traditional separate models would miss.